EXPLICIT PERMUTATIONS OF $(\mathbb{Z}/4\mathbb{Z})^m$ WITH UNIFORMLY BOUNDED DIFFERENTIAL UNIFORMITY

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Abstract. Symmetric primitives over Galois rings need nonlinear components whose differentials with respect to ring addition are controlled. Let $R = \mathrm{GR}(4, m)$ and $q = 2^m$, so that $(R, +) \cong (\mathbb{Z}/4\mathbb{Z})^m$ has order $N = q^2$. For every $m \geq 1$ we construct an explicit permutation G_c of R all of whose differentials with a nonzero input difference have probability at most $36/N$, whereas every permutation polynomial of R has such a differential of probability at least $2/\sqrt{N}$ (Rønjom–Sandrib). The bound holds for the fixed permutation itself, not on average over keys. Equivalently, $\Delta(G_c) \leq 36$ against $\Delta \geq 2\sqrt{N}$ for permutation polynomials; since every permutation has $\Delta \geq 2$, the least differential uniformity of a permutation of $(\mathbb{Z}/4\mathbb{Z})^m$ lies between 2 and 36 for every m. In Teichmüller digits $(a, b) \in \mathbb{F}_q^2$ and for $c \in \mathbb{F}_q$ of absolute trace 1,

$$G_c(a, b) = \big(L,\ (L^2 + L + c)\, a\big), \qquad L = (a^2 + a + c)\, b.$$

The map is $F_c \circ J \circ F_c$, where $F_c(a, b) = (a, (a^2 + a + c)b)$ is induced by a polynomial over R and J exchanges the digits: a single digit exchange between two polynomial layers yields a bound independent of m. The proof reduces the difference equations, in each of three types of directions, to one-variable polynomials whose degrees do not depend on m. For general maps $F \circ J \circ F$ with $F(a, b) = (p(a), h(a) + d(a)b)$, where p, d, h have degrees bounded independently of m, $p' \neq 0$ and d has no zeros, a Jacobian identity in characteristic 2 bounds the differentials with nonzero low input and output differences independently of m; the remaining directions need a separate argument, which we give for an explicit family ($\Delta \leq 440$, resp. 510, for odd m).

1. Introduction

For a map G on a finite abelian group $(R, +)$ put $\delta_G(\alpha, \beta) = \#\{x \in R : G(x + \alpha) - G(x) = \beta\}$ and $\Delta(G) = \max_{\alpha \neq 0, \beta} \delta_G(\alpha, \beta)$. Over $\mathbb{F}_2^n$ the group is elementary abelian, and power maps such as x^3 are permutations with $\Delta = 2$ for every odd n. Symmetric primitives over the integer rings $\mathbb{Z}_{2^k}$ and over Galois rings, studied for use in multiparty computation and homomorphic encryption [5, 3], raise the same question for groups that are not elementary abelian. The simplest family of this kind is $(\mathbb{Z}/4\mathbb{Z})^m$, the additive group of $R = \mathrm{GR}(4, m)$, and there polynomial constructions face a restriction: Rønjom and Sandrib proved that every permutation polynomial of $\mathrm{GR}(p^k, m)$ has $\Delta \geq 2p^{m(k-1)}$ [1, Theorem 3] (every polynomial function has $\Delta \geq p^{m(k-1)}$ [1, Proposition 2]); for $\mathrm{GR}(4, m)$ this is $\Delta \geq 2q = 2\sqrt{|R|}$, so a permutation with smaller Δ cannot be a polynomial function. Splitting an element into digits, applying maps to the digits and recombining is an established design strategy. It appears in the "cut-and-sew" S-boxes over $\mathbb{Z}_q$ for composite q [5, §4.3], where for $q = p^2$ the two digits are also exchanged; in the Kintsugi strategy over prime fields [6, §2]; and as "digit manipulation" over $\mathbb{Z}_{p^n}$ in Beyne, Grassi, Øygarden, Richterová and Sandrib [3, §9]. These constructions act on cyclic groups $\mathbb{Z}_q$ or on $\mathbb{F}_p$. Here we consider the groups $(\mathbb{Z}/4\mathbb{Z})^m$, noncyclic for $m > 1$, and give an explicit family whose differential uniformity is bounded independently of m. Rønjom and Sandrib also classify the table-defined (non-polynomial) permutations of $\mathrm{GR}(4, 2)$ with $\Delta = 2$ [1, §5]; no family covering all m is given there. On cyclic groups $\mathbb{Z}_n$ several infinite families of permutations with $\Delta = 2$ are known (see [2] and the references

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there), but $\mathbb{Z}_{4^m}$ and $(\mathbb{Z}/4\mathbb{Z})^m$ are different groups for $m > 1$. Permutation polynomials are subject to the lower bound $\Delta \geq 2\sqrt{|R|}$ [1], and the one-sided digit-exchange maps $J \circ F$ of [4] attain $\Delta = q = \sqrt{|R|}$, which is optimal within that class (Theorem 3.1). The family constructed here has $\Delta \leq 36$ for every m, so the optimal order is constant.

Theorem 1.1. *Let $m \geq 1$, $q = 2^m$, and $c \in \mathbb{F}_q$ with $\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_2}(c) = 1$. In Teichmüller digits, the map $G_c = F_c \circ J \circ F_c$ with $F_c(a, b) = (a, D(a)b)$, $D(x) = x^2 + x + c$, $J(a, b) = (b, a)$, is a permutation of* $\mathrm{GR}(4, m)$ *and*

$$\Delta(G_c) \leq 36.$$

Let D_m be the least Δ over all permutations of $(\mathbb{Z}/4\mathbb{Z})^m$. Theorem 1.1 and Proposition 3.2 give $2 \leq D_m \leq 36$ for every m, so $D_m = \Theta(1)$, whereas every permutation polynomial has $\Delta \geq 2\sqrt{|R|}$. In terms of probabilities, with $N = |R| = q^2$, every differential of G_c with a nonzero input difference has probability at most $36/N$, which is of the optimal order N^{-1}. For example, for $m = 8$ the ring R has 65,536 elements (a 16-bit state): every permutation polynomial of R has a differential of probability at least $512/65{,}536 = 1/128$, while every differential of G_c with a nonzero input difference has probability at most $36/65{,}536$, more than 14 times smaller. In the instances of Table 1 ($m \leq 10$) the exact values of $\Delta(G_c)$ are at most 20. Since F_c is itself induced by a polynomial (Section 2.2), the polynomial lower bound is overcome by a single digit exchange between two polynomial layers. The proof (Section 4) is elementary, and the degrees it uses do not depend on m.

Comparison. With $F_{p,d,h}(a, b) = (p(a), h(a) + d(a)b)$:

map	digit system	Δ
permutation polynomial	—	$\geq 2q$ [1]
$J \circ F$ (output exchange only)	any	$\geq q$ (Thm. 3.1)
$F_c \circ J \circ F_c$	Teichmüller	≤ 36 for all m (Thm. 1.1)
$F \circ J \circ F$, $p = x^3$, $d = x^4 + x^2 + 1$, $h \in \{0, x^2, x^3, x^5\}$, m odd	Teichmüller	≤ 440, resp. 510 (Thm. 5.4)
literal exchange J ($= F \circ J \circ F$ with $p = x$, $d = 1$, $h = 0$)	Teichmüller	$= q$ (Remark 5.3)
literal exchange J	coordinate	$= q^2/2$ (Prop. 6.2)
$F \circ J \circ F$, 1-periodic h, d	coordinate	$\geq q/2$ (Prop. 6.1)

The class $J \circ F$ in the second line contains the maps studied in [4] (coordinate digits, p APN, d affine), for which $\Delta = q$ is attained; Theorem 3.1 shows that q cannot be improved within this class. Unlike [4], which also gives key-averaged and multi-round statements, we consider a single fixed permutation and all its differentials. For the families of Theorems 1.1 and 5.4, one more polynomial layer after the exchange removes the dependence on m.

Scope. We do not determine D_m, and we do not consider other attacks (linear, integral, algebraic) or implementation cost. Open questions are collected in Section 8.

2. Setting

2.1. **Digits.** $R = \mathrm{GR}(4, m) = W_2(\mathbb{F}_q)$. Every $x \in R$ is uniquely $x = \mathrm{T}(a) + 2\mathrm{T}(b)$ with $a, b \in \mathbb{F}_q$, where T is the Teichmüller lift; we write $x = (a, b)$. The identity $\mathrm{T}(a) + \mathrm{T}(u) = \mathrm{T}(a+u) + 2\mathrm{T}(\sqrt{au})$ gives

$$(a, b) + (u, v) = (a + u,\ b + v + \sqrt{au}), \qquad -(a, b) = (a,\ a + b), \tag{1}$$

hence

$$(y_0, y_1) - (z_0, z_1) = (y_0 + z_0,\ y_1 + z_0 + z_1 + \sqrt{y_0 z_0}). \tag{2}$$

The *coordinate* digit system fixes a basis $e_0 = 1, e_1, \dots, e_{m-1}$ of $\mathbb{F}_q$ over $\mathbb{F}_2$ whose first vector is 1 (for definiteness, the polynomial basis $1, \omega, \dots, \omega^{m-1}$ of $\mathbb{F}_q = \mathbb{F}_2(\omega)$) and lifts coordinates to $\{0,1\} \subset \mathbb{Z}/4$. The resulting map depends on the basis; all statements below about this system are for a basis with $e_0 = 1$. Then (1) holds with the carry $\sqrt{au}$ replaced by the coordinatewise product $a \odot u$.

We apply the same formulas in both digit systems; they give different permutations of R. The change of digits has the form $C(a,b) = (a, b + \gamma(a))$, under which $F_{p,d,h}$ becomes $F_{p,d,h'}$ with $h'(a) = h(a) + d(a)\gamma(a) + \gamma(p(a))$, and CJC^{-1} is in general not the literal exchange J.

2.2. **The maps.** For $p, d, h \in \mathbb{F}_q[x]$ with p a permutation of $\mathbb{F}_q$ and d without zeros in $\mathbb{F}_q$, $F = F_{p,d,h}$ is a permutation of $\mathbb{F}_q^2$, hence of R; so are $J \circ F$ and $F \circ J \circ F$. Theorem 1.1 is the case $p = x$, $h = 0$, $d = x^2 + x + c$.

We call $F_{p,d,h}$ a polynomial layer because it is induced by a polynomial over R. For $\mathsf{F} \in R[X]$ with reduction $\bar{\mathsf{F}} \in \mathbb{F}_q[X]$, the identity $\mathsf{F}(y + 2z) = \mathsf{F}(y) + 2z\,\mathsf{F}'(y)$ shows that F acts on digits as $(a,b) \mapsto (\bar{\mathsf{F}}(a),\ h_0(a) + \bar{\mathsf{F}}'(a)\,b)$, where $h_0(a)$ is the high digit of F at $(a,0)$; conversely, every triple of functions p, d, h arises in this way, by interpolation with the factor $X^q - X$ and a correction $2\hat{H}$ of the high digit. This is [4, Proposition "Coordinate form and realizability"], stated there for coordinate digits; the argument uses only that $2w \in R$ depends on w mod 2 alone, and holds verbatim for Teichmüller digits. The exchange J is not of this form. Since every permutation polynomial of R has $\Delta \geq 2q$ [1], Theorem 1.1 shows that for $m \geq 5$ the permutation G_c is not induced by any polynomial over R.

3. Output exchange alone: $\Delta \geq q$

Theorem 3.1. *Let $p, h : \mathbb{F}_q \to \mathbb{F}_q$ be arbitrary and $d : \mathbb{F}_q \to \mathbb{F}_q^\times$. In either digit system, $\Delta(J \circ F_{p,d,h}) \geq q$.*

Proof. Teichmüller digits: $B = J \circ F$ is $B(a,b) = (h(a) + d(a)b,\ p(a))$. Since $|\mathbb{F}_q^\times| < q$, d is not injective; choose a_0 and $u \neq 0$ with $d(a_0 + u) = d(a_0)$, and v with $h(a_0 + u) + h(a_0) + d(a_0)(v + \sqrt{a_0 u}) = 0$. For $x = (a_0, b)$ and $\alpha = (u, v)$ the low digits of $B(x + \alpha)$ and $B(x)$ coincide, so by (2) $B(x+\alpha) - B(x) = (0,\ p(a_0 + u) + p(a_0))$ for all q values of b. The coordinate case is identical with $a_0 \odot u$ in place of $\sqrt{a_0 u}$ (for the maps of [4], $\Delta(B) = q$ exactly). □

Proposition 3.2. *Every map $G : (\mathbb{Z}/4\mathbb{Z})^m \to (\mathbb{Z}/4\mathbb{Z})^m$, $m \geq 1$, has $\Delta(G) \geq 2$. Consequently, for every m, $2 \leq \min_G \Delta(G) \leq 36$, the minimum over all permutations.*

Proof. Let $\alpha \neq 0$ with $2\alpha = 0$ and put $D(x) = G(x+\alpha) - G(x)$. Then $D(x+\alpha) = G(x) - G(x+\alpha) = -D(x)$. If some x has $D(x) = \beta$ with $2\beta = 0$, then $D(x+\alpha) = \beta$ as well, so $\delta_G(\alpha, \beta) \geq 2$. Otherwise D misses the $2^m \geq 2$ elements of order at most 2, so D is not surjective and some value is taken twice. □

For permutations the bound $\Delta \geq 2$ is immediate in any finite group of order > 2, since $x \mapsto G(x + \alpha) - G(x)$ never takes the value 0; Proposition 3.2 extends it to arbitrary maps.

4. Proof of Theorem 1.1

Permutation. The image of $x \mapsto x^2 + x$ on $\mathbb{F}_q$ is the trace-zero hyperplane, so $\mathrm{Tr}(c) = 1$ gives $D(a) \neq 0$ for all a; F_c multiplies b by a nonzero scalar on each fibre and is bijective. Explicitly

$$G_c(a,b) = \big(L,\ D(L)\,a\big), \qquad L = D(a)\,b.$$

Difference equations. Fix $\alpha = (u,v) \neq 0$ and $\beta = (s,t)$, and write $a = A^2$, $u = U^2$ (Frobenius is bijective on $\mathbb{F}_q$, so A runs over $\mathbb{F}_q$ and $\sqrt{au} = AU$). By (1), $x + \alpha = (a + u,\ b + v + AU)$, so

$$L = D(a)\,b, \qquad L' = D(a+u)\,(b + v + AU)$$

are the low digits of $G_c(x)$ and $G_c(x+\alpha)$. By (2), $G_c(x+\alpha)-G_c(x)=\beta$ is equivalent to

$$L'+L=s, \tag{3}$$

$$t=D(L')(a+u)+L+D(L)\,a+\sqrt{LL'}. \tag{4}$$

Given (3), $D(L')=D(L)+s^2+s$ and $\sqrt{LL'}=\sqrt{L^2+sL}=L+\sqrt{sL}$, so (4) becomes $t=u\,D(L)+(s^2+s)(a+u)+\sqrt{sL}$, and squaring (Frobenius is injective) gives the equivalent condition

$$u^2\big(L^4+L^2+c^2\big)+(s^4+s^2)(a^2+u^2)+sL+t^2=0. \tag{5}$$

We count pairs $(A,b)\in\mathbb{F}_q^2$ satisfying (3) and (5); the map $(A,b)\mapsto x=(A^2,b)$ is a bijection.

Case A: $u\notin\{0,1\}$. Put $\kappa=u^2+u\neq 0$, so $D(a+u)=D(a)+\kappa$. Then (3) reads $\kappa b+D(a+u)(v+AU)+s=0$, which determines b from A, and

$$L=N(A)/\kappa,\qquad N(A)=D(a)\,s+D(a)D(a+u)(v+AU),$$

a polynomial in A of degree 9 with leading coefficient U. Substituting into (5) and multiplying by κ^4 gives

$$u^2\big(N^4+\kappa^2N^2+c^2\kappa^4\big)+(s^4+s^2)(A^4+u^2)\kappa^4+sN\kappa^3+t^2\kappa^4=0,$$

a polynomial in A of degree 36 with leading coefficient $u^2U^4=U^8\neq 0$ (all other terms have degree at most 18). Hence at most 36 values of A, each with one b: $\delta\le 36$. This includes $s=0$.

Case B: $u=1$. Then $U=1$ and $D(a+1)=D(a)$, so (3) reads $D(A^2)(v+A)+s=0$, a monic polynomial of degree 5 in A: at most 5 values. For each, (5) is a monic polynomial of degree 4 in L, and $b=L/D(a)$. Hence $\delta\le 20$.

Case C: $u=0$, $v\neq 0$. Then (3) reads $v\,D(a)=s$; there is no solution unless $s\neq 0$, and then $D(a)=s/v$ has at most 2 roots a. For each, (5) becomes $(s^4+s^2)a^2+sL+t^2=0$, linear in L with coefficient $s\neq 0$. Hence $\delta\le 2$.

The three cases cover all $\alpha\neq 0$, so $\Delta(G_c)\le 36$. □

Remark 4.1. The argument holds for all m and uses neither an APN property nor intersection theory. The constant comes from Case A: the degree $36=4\cdot 9$ of u^2N^4. In the instances of Table 1 the observed values are at most 20 and are not monotone in m (10 for $m=5$, 9 for $m=6$).

Remark 4.2 (Evaluation)**.** On Teichmüller digits, G_c costs two general multiplications, two squarings and four additions in $\mathbb{F}_q$; given the output (L,y), the inverse $a=y/D(L)$, $b=L/D(a)$ costs two inversions, two multiplications and two squarings. Passing between $x\in R$ and its digits requires $a=x \bmod 2$ and $\mathrm{T}(a)$, i.e. m squarings in R (since $\mathrm{T}(a)=\tilde{a}^{\,q}$ for any lift $\tilde{a}$). In settings where elements of R are held as shares modulo 4, the reduction $x \bmod 2$ is a bit extraction whose cost depends on the protocol; we do not analyse it here.

5. A general method for $F\circ J\circ F$

For general (p,d,h) the low-digit equation is no longer linear in b, and a uniform bound needs an argument about common components. Let $G=F\circ J\circ F$, $\alpha=(u,v)\neq 0$, $\beta=(s,t)$, $a=A^2$, $u=U^2$, and

$$L=h(a)+d(a)b,\quad L'=h(a+u)+d(a+u)(b+v+AU),$$
$$Z_0=p(L),\ Z_1=h(L)+d(L)p(a),\quad Y_0=p(L'),\ Y_1=h(L')+d(L')p(a+u).$$

By (2), after squaring the high digit,

$$\delta_G(\alpha,\beta)=\#\{(A,b)\in\mathbb{F}_q^2:\ E_1=E_2=0\},\quad E_1=Y_0+Z_0+s,\quad E_2=(Y_1+Z_0+Z_1+t)^2+Y_0Z_0. \tag{6}$$

Lemma 5.1. *Put $\Psi=Y_0+Z_0$ and $\Phi=(Y_1+Z_0+Z_1)^2+Y_0Z_0$. Then $\det\partial(\Psi,\Phi)/\partial(A,b)=\Psi\cdot p'(L)\,p'(L')\cdot d(a)\,d(a+u)\cdot U$.*

Proof. In characteristic 2 squares have zero derivative, so Φ may be replaced by Y_0Z_0. The pair (Ψ, Y_0Z_0) consists of the elementary symmetric functions of (Z_0, Y_0), with Jacobian $Z_0 + Y_0 = \Psi$; next $\partial(Z_0, Y_0)/\partial(L, L') = p'(L)p'(L')$. Since L involves A only through $a = A^2$, $\partial L/\partial A = 0$, $\partial L/\partial b = d(a)$, $\partial L'/\partial b = d(a+u)$, $\partial L'/\partial A = d(a+u)U$, so $\partial(L, L')/\partial(A, b) = d(a)d(a+u)U$. □

Theorem 5.2. *Let $p, d, h \in \mathbb{F}_q[x]$ with $p' \neq 0$ and d without zeros in $\mathbb{F}_q$, and let $K = \overline{\mathbb{F}}_2$. If $u \neq 0$ and $s \neq 0$, every common component of E_1 and E_2 in K^2 is a line $A = c_0$ with $c_0 \notin \mathbb{F}_q$, and $\delta_G(\alpha, \beta) \leq \deg E_1 \cdot \deg E_2$.*

Proof. Let Γ be an irreducible common component. On Γ, $\Psi \equiv s$ and $\Phi \equiv t^2$, so both differentials vanish on the tangent line at smooth points and the Jacobian of Lemma 5.1 vanishes on Γ. As $\Psi = s \neq 0$ and $U \neq 0$, Γ lies in $\{p'(L) = 0\}$, $\{p'(L') = 0\}$, $\{d(a) = 0\}$ or $\{d(a+u) = 0\}$. The last two are unions of lines $A = c_0$ with c_0^2 or $c_0^2 + u$ a root of d, so $c_0 \notin \mathbb{F}_q$.

Suppose $p'(L) = 0$ on Γ, so $L \equiv r$ for a root r of p'. If Γ is vertical, $A \equiv c_0$ while b varies along Γ, so $L = h(c_0^2) + d(c_0^2)b \equiv r$ forces $d(c_0^2) = 0$ and $c_0 \notin \mathbb{F}_q$. If Γ is not vertical, A is transcendental over K on Γ and $b = (r - h(A^2))/d(A^2)$, so the function field is $K(\Gamma) = K(A)$ and

$$L'|_\Gamma = E(A^2) + U\, d(A^2 + u)\, A, \qquad E \in K(A^2).$$

Since $K(A) = K(A^2) \oplus A\, K(A^2)$ and $U\, d(A^2+u) \neq 0$, $L'|_\Gamma \notin K$; hence $\Psi|_\Gamma = p(r) + p(L'|_\Gamma)$ is not constant, contradicting $\Psi \equiv s$. The case $p'(L') = 0$ is symmetric: then $L|_\Gamma = E'(A^2) + U\, d(A^2)\, A$ with $E' \in K(A^2)$.

Thus $C = \gcd(E_1, E_2)$ is a product of vertical lines without $\mathbb{F}_q$-points, every $\mathbb{F}_q$-solution is an isolated point of $V(E_1/C) \cap V(E_2/C)$, and affine Bézout gives the bound. □

The second-stage high digit enters E_2 only inside the square, so Theorem 5.2 holds for every h. The bound depends on m only through the degrees of p, d, h, and is uniform in m when these degrees are bounded.

Remark 5.3. Theorem 5.2 bounds $\delta_G(\alpha, \beta)$ only for $u \neq 0$, $s \neq 0$, and assumes $p' \neq 0$. The directions with $u = 0$ or $s = 0$ need a separate argument, and without one the differential uniformity of $F \circ J \circ F$ need not be bounded independently of m. For example, $p = x$, $d = 1$, $h = 0$ satisfy the hypotheses, but $F = \mathrm{id}$ and $F \circ J \circ F = J$. For $\alpha = (1, 0)$ and $x = (0, b)$ we have $x + \alpha = (1, b)$, and by (2)

$$J(x + \alpha) - J(x) = (b, 1) - (b, 0) = (0, 1) \qquad \text{for all } b \in \mathbb{F}_q,$$

so $\Delta(J) \geq q$. In fact $\Delta(J) = q$: for $\alpha = (u, v)$ put $s = v + \sqrt{au}$; then $J(x+\alpha) - J(x) = (s,\, u + \sqrt{bs})$. If $u \neq 0$, the value s determines a; then t determines b when $s \neq 0$, and when $s = 0$ we have $t = u$ with b free. If $u = 0$, then $s = v \neq 0$, t determines b, and a is free. The same row $\alpha = (1, 0)$ gives $\Delta \geq q$ for the literal exchange in the coordinate system, but its exact value there is $q^2/2$ (Proposition 6.2); the two literal exchanges are different permutations of R. In Theorem 1.1 and Theorem 5.4 the remaining directions are treated by hand.

Theorem 5.4. *Let m be odd, $p = x^3$, $d = x^4 + x^2 + 1$, $h \in \{0, x^2, x^3, x^5\}$, and $G = F \circ J \circ F$ in Teichmüller digits. Then $\Delta(G) \leq 440$ for $h \in \{0, x^2, x^3\}$ and $\Delta(G) \leq 510$ for $h = x^5$.*

Proof. For m odd, p is bijective and $d = (x^2 + x + 1)^2$ has no zeros in $\mathbb{F}_q$.

$u \neq 0$, $s \neq 0$. By Theorem 5.2 the $\mathbb{F}_q$-solutions are isolated roots in K^2 of a system whose supports are contained in the generic supports of (E_1, E_2); the total degrees are at most $(27, 84)$, resp. $(29, 98)$, which already gives 2268, resp. 2842. These supports contain the origin and a pure power of each variable, so the pair is cornered, and it is nice (e.g. $(A^{27} + 1, b^8 + 1)$, resp. $(A^{29} + 1, b^{10} + 1)$, has finitely many common roots). Rojas' Affine Point Theorem II, valid over any algebraically closed field, bounds the number of isolated affine roots, with multiplicity, of

every system with these supports by the mixed volume of the Newton polygons [7, Affine Point Theorem II]; this applies to each specialised system. The Newton polygons are

$$P_1 = \mathrm{conv}\{(0,0),(27,0),(25,2),(16,3),(0,3)\},$$
$$P_2 = \mathrm{conv}\{(0,0),(84,0),(72,8),(0,8)\} \quad (\mathrm{MV} = 440),$$
$$P_1 = \mathrm{conv}\{(0,0),(29,0),(25,2),(16,3),(0,3)\},$$
$$P_2 = \mathrm{conv}\{(0,0),(98,0),(82,8),(64,10),(0,10)\} \quad (\mathrm{MV} = 510),$$

the second pair for $h = x^5$.

$u \neq 0$, $s = 0$. On $\mathbb{F}_q$-points $p(L) = p(L')$ forces $L = L'$, and then h cancels: $E_2 \iff t = u\,d(L)(a^2 + ua + u^2)$. If $u \notin \{0,1\}$, put $c_1 = u^4 + u^2 \neq 0$; then $L = L'$ gives $L = N/c_1$ with $N = d(a)h(a+u) + d(a+u)h(a) + d(a)d(a+u)(v + AU)$, of degree 17 in A with leading coefficient U, and the remaining equation is non-constant of degree 72 in A; b is then determined. If $u = 1$, then $d(a+1) = d(a)$, $L' = L$ reads $h(a+1) + h(a) + d(a)(v + A) = 0$, whose odd part $d(a)A$ is nonzero, so at most 9 values of A; for each, $d(L) = t/(a^2 + a + 1)$ and $L \mapsto L^4 + L^2$ is two-to-one. Hence $\delta \leq 72$.

$u = 0$, $v \neq 0$. Put $w = v\,d(a) \neq 0$; in the coordinates (a, L) (a bijection of $\mathbb{F}_q^2$), $E_1 = wL^2 + w^2L + w^3 + s$. If $s = 0$ then $L^3 = (L + w)^3$, impossible as cubing is bijective and $w \neq 0$; so $s \neq 0$. Reducing E_2 modulo E_1 over $K(a)$ leaves a polynomial of degree 1 in L with coefficient $s^2/w \neq 0$, so each a gives at most one L. Eliminating L, $\mathrm{Res}_L(E_1, E_2) = d(a)^2 g(a)$ (resp. $d(a)^4 g(a)$ for $h = x^5$) with g of degree 88 (resp. 96), nonzero leading coefficient for $v \neq 0$, and only even powers of a. As $\mathbb{F}_q$ is perfect, g is the square of a polynomial of degree 44 (resp. 48), so $\delta \leq 44$ (resp. 48). □

Remark 5.5. For the doubly exchanged maps $J \circ F \circ J \circ F$ of the same family, an analogous Jacobian identity holds, $\det \partial(Y_1 + Z_1, Y_1 Z_1)/\partial(A, b) = (Y_1 + Z_1)\, g(L) g(L')\, d(a) d(a+u) U$ with $g = h' + d' \cdot p$; whether these maps also have Δ bounded independently of m is left open (Section 8).

6. The coordinate digit system

Proposition 6.1. *In the coordinate digit system with $e_0 = 1$, if $d(a+1) = d(a)$ and $h(a+1) = h(a)$ for all a (for instance $h = 0$, $d = x^4 + x^2 + 1$, or $h = 0$, $d = x^2 + x + c$), then $\Delta(F \circ J \circ F) \geq q/2$.*

Proof. Take $\alpha = (1, 0)$. Since $1 = e_0$, the carry is $a \odot 1 = a_0$, the first coordinate of a. For the $q/2$ values of a with $a_0 = 0$ and all b, $x + \alpha = (a + 1, b)$ and $L' = L$, so $G(x + \alpha) - G(x) = \big(0,\ d(L)(p(a+1) + p(a))\big)$. These $q^2/2$ inputs land in at most q values of β. □

For $h = 0$ and $d = x^4 + x^2 + 1$ (with $p = x^3$) the row $\alpha = (1, 0)$ alone reaches $6, 22, 74$ for $m = 3, 5, 7$. In general the kernel of $a \mapsto a \odot u$ has $2^{m - \mathrm{wt}(u)}$ elements; in the Teichmüller system, by contrast, $a \mapsto \sqrt{au}$ is injective for $u \neq 0$. In the coordinate system the carry is a linearised polynomial of degree up to 2^{m-1}, so the bounded-degree arguments above do not apply; whether $F \circ J \circ F$ has bounded Δ there for some h that is not 1-periodic is open.

Proposition 6.2. *In the coordinate digit system with $e_0 = 1$, the literal exchange $J_\sigma(a, b) = (b, a)$ has $\Delta(J_\sigma) = q^2/2$.*

Proof. For $\alpha = (u, v)$ put $s = v + a \odot u$. Then $J_\sigma(x + \alpha) - J_\sigma(x) = (s,\ u + b \odot s)$. For a target (s, t), the set of a with $a \odot u = s + v$ has 0 or $2^{m - \mathrm{wt}(u)}$ elements, and the set of b with $b \odot s = t + u$ has 0 or $2^{m - \mathrm{wt}(s)}$ elements; hence $\delta(\alpha, (s, t)) \leq 2^{2m - \mathrm{wt}(u) - \mathrm{wt}(s)}$. If $u \neq 0$ then $\mathrm{wt}(u) \geq 1$; if $u = 0$ then $s = v \neq 0$ and $\mathrm{wt}(s) \geq 1$. In both cases $\mathrm{wt}(u) + \mathrm{wt}(s) \geq 1$, so $\delta \leq q^2/2$. For $\alpha = (0, e_0)$ and $\beta = (e_0, 0)$, every a and every b with $b_0 = 0$ is a solution, so the bound is attained. □

For $m = 1$ both literal exchanges have $\Delta = 2$; for $m > 1$ they differ.

7. Data

TABLE 1. Exact $\Delta(G_c)$ for the tested parameters. Ring model: $\mathrm{GR}(4,m) = \mathbb{Z}_4[X]/(f)$ with ring addition only and digits recovered by table lookup, for all c with $\mathrm{Tr}(c) = 1$ when $m \le 5$ and for two values of c when $m = 6$. Digit formulas (1)–(2) (all $N-1$ directions, with a coverage check): one value of c for each $m \le 10$ and two for $m = 8$; these agree with the ring model for $m \le 6$. The field polynomial and the values of c are listed in the supplementary material. For each m, the value was the same for every tested c.

m	1	2	3	4	5	6	7	8	9	10
q	2	4	8	16	32	64	128	256	512	1024
$\Delta(G_c)$	2	4	4	6	10	9	12	14	16	20

8. Open problems

(1) Determine D_m. By Table 1 and [1, §5], $D_1 = D_2 = 2$. Are there permutations of $(\mathbb{Z}/4\mathbb{Z})^m$ with $\Delta = 2$ for every m?
(2) How far can the constant 36 be lowered, for G_c or for other explicit families? For the tested parameters with $m \le 10$ (Table 1) the values of $\Delta(G_c)$ are at most 20.
(3) For families (p, d, h) of bounded degree, Theorem 5.2 bounds the differentials with $u \neq 0$, $s \neq 0$ independently of m. For which such families are the remaining directions also bounded independently of m? Does the same hold for the doubly exchanged maps $J \circ F \circ J \circ F$?
(4) Can $\Delta(F \circ J \circ F)$ remain bounded independently of m in the coordinate digit system for some h that is not 1-periodic (Section 6)?

Appendix A. Computational checks

The proofs above are complete without computation. The following scripts (supplementary material; Magma, and one C program with its runner) check the stated identities and degrees with `assert`, and recompute the table.

- `t4_v2_bounded36_symbolic.m`: Theorem 1.1, from the unreduced equations (3)–(4): in Case A, N of degree 9 with leading coefficient U and the eliminated equation of degree 36 with leading coefficient U^8/κ^4; in Case B, the low equation monic of degree 5 in A and the high equation monic of degree 4 in L; in Case C, the low equation $vD(a) + s$ and the high equation linear in L with coefficient s.
- `t5_ringmodel_v2.m` (parameters `mlo`, `mhi`, `cmax`): $\Delta(G_c)$ in the ring model, using ring addition only.
- `gc_ddt_v2.c` with `run_gc_v2.sh`: $\Delta(G_c)$ from the digit formulas, in parallel shards; the runner accepts a run only if every shard exits with status 0 and the recorded number and sum of directions of each shard equal those of its residue class $\alpha \equiv k \pmod K$.
- `c1_form1_assert.m`: Lemma 5.1 as a polynomial identity (the family of Theorem 5.4 and random (p, d, h)), and the mixed volumes 440 and 510.
- `t3_u0_form1_v2.m`: the $u = 0$ elimination in Theorem 5.4 (exponent of the removed factor $d(a)$, degree and leading coefficient of g, even exponents, and the linear remainder).
- `t1_J_counterexample.m`: $\Delta(J) = q$ in the ring model for $m \le 5$ (Teichmüller digits, Remark 5.3); `r05_t1_digit_swap.m`: both literal exchanges for $m \le 5$, with $\Delta = q$ (Teichmüller) and $\Delta = q^2/2$ (coordinate, Proposition 6.2).
- `t6_sigma_row.m`: the row $\alpha = (1, 0)$ in the coordinate system (Proposition 6.1).

The supplementary material contains the scripts, their raw outputs, a manifest with SHA-256 hashes, and instructions for reproducing each run. It is archived with the paper at `https://doi.org/10.5281/zenodo.23875215` and at `https://github.com/hirokifukui/z4m-bounded-differential-uniformity`.

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